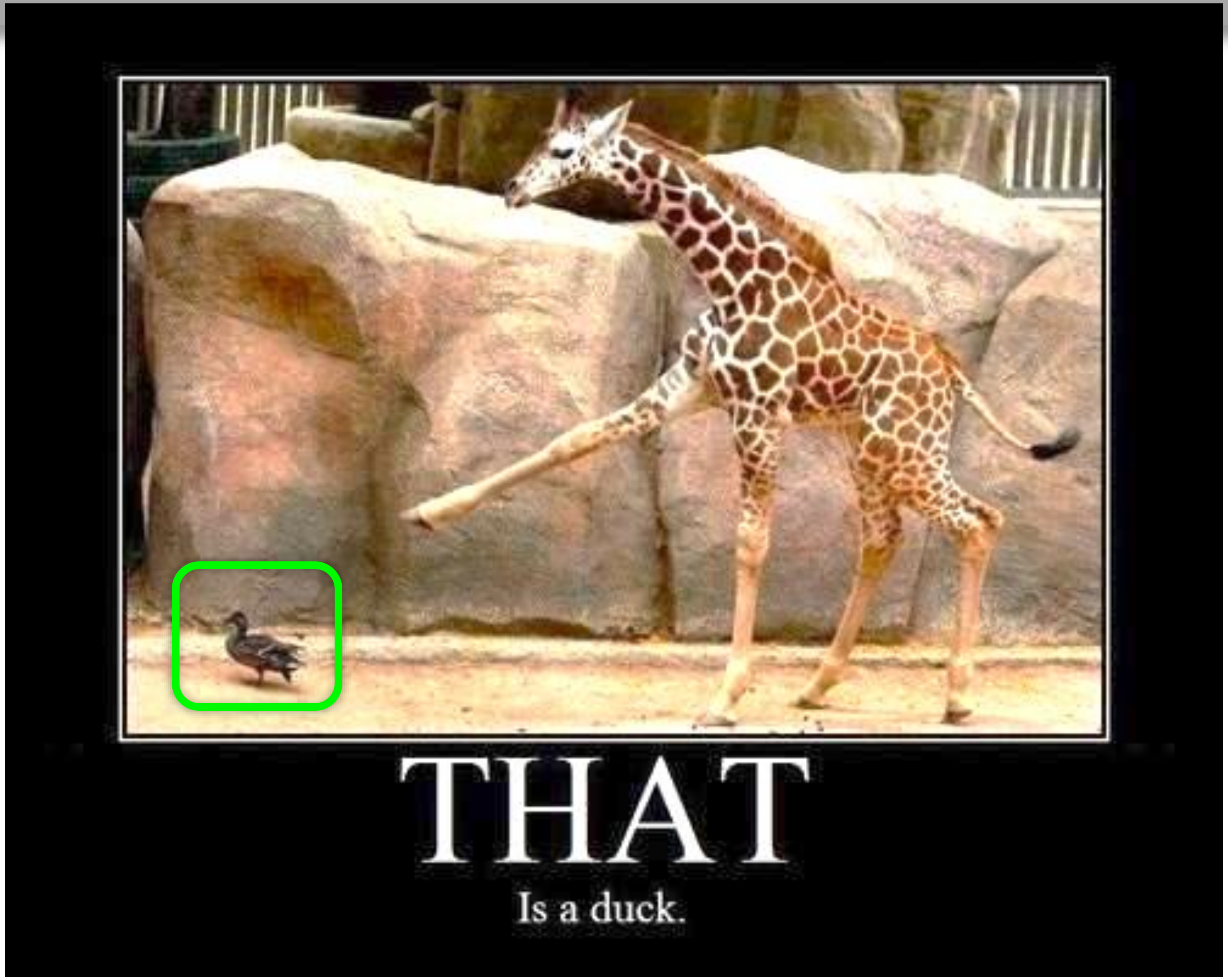
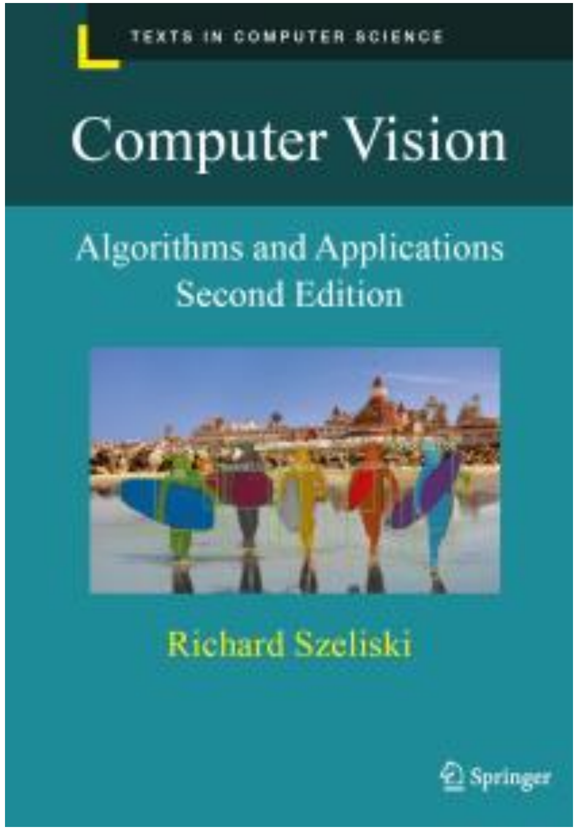
Computer Vision

Introduction to Recognition



Important information



Textbook

Rick Szeliski, *Computer Vision: Algorithms and Applications* online at: http://szeliski.org/Book/

Many of the slides in this course are modified from the excellent class notes of similar courses offered in other schools by Noah Snavely, Prof Yung-Yu Chuang, Fredo Durand, Alyosha Efros, Bill Freeman, James Hays, Svetlana Lazebnik, Andrej Karpathy, Fei-Fei Li, Srinivasa Narasimhan, Silvio Savarese, Steve Seitz, Richard Szeliski, and Li Zhang. The instructor is extremely thankful to the researchers for making their notes available online. Please feel free to use and modify any of the slides, but acknowledge the original sources where appropriate.

All readings are from Richard Szeliski, Computer Vision: Algorithms and Applications, 2nd Edition, unless otherwise noted.

Where we go from here

- What we know: Geometry
  - What is the shape of the world?
  - How does that shape appear in images?
  - How can we infer that shape from one or more images?

- What's next: Recognition
  - What are we looking at?
  - **Representations of visual content**

- Generative models

What is “Recognition”?

Next few slides adapted from Li, Fergus, & Torralba’s excellent short course on category and object recognition

What is “Recognition”?

- Verification: is that a lamp?

What is “Recognition”?

- Verification: is that a lamp?
- Detection: where are the people?

What is “Recognition”?

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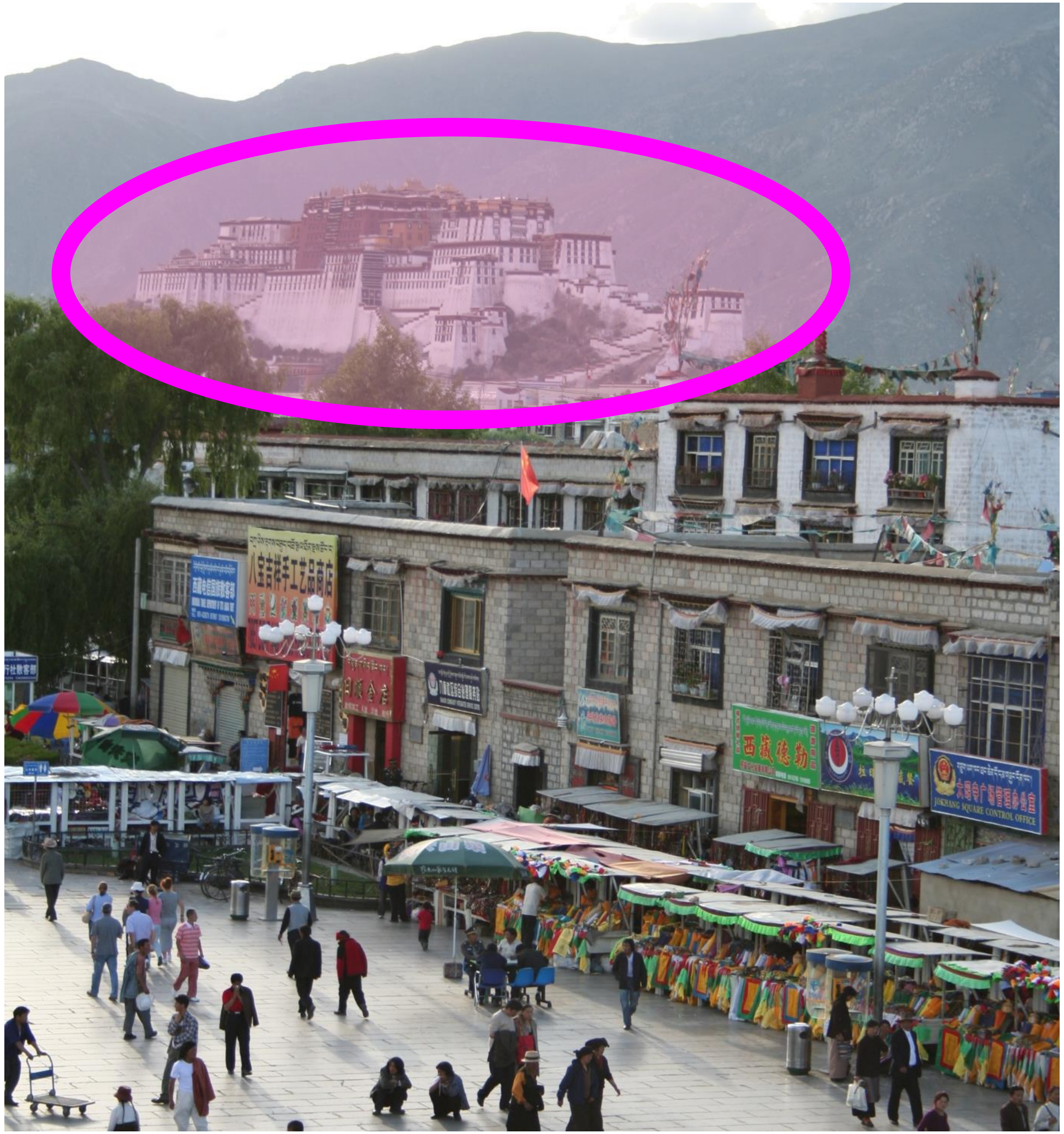
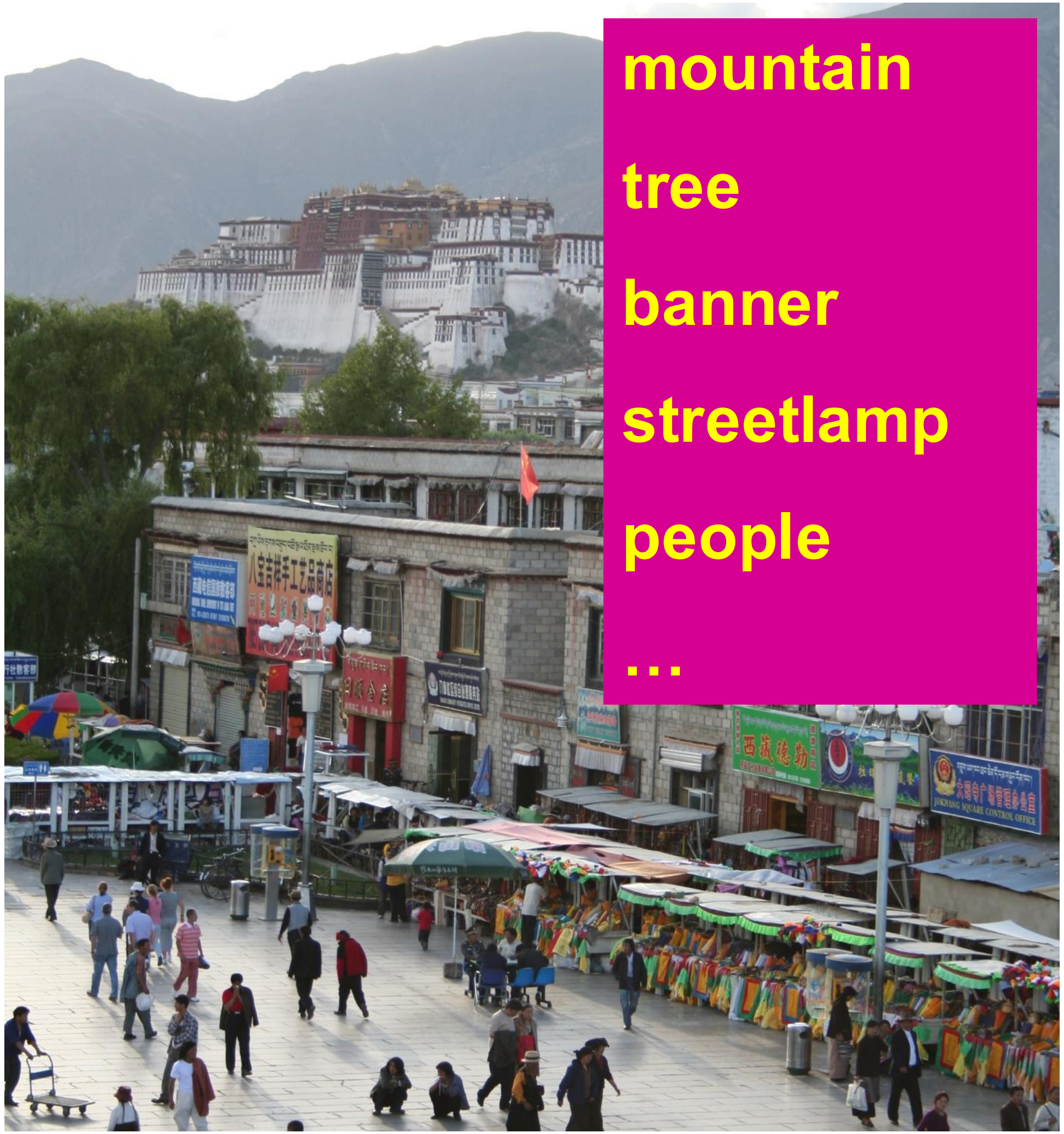
- Detection: where are the people?
- Identification: is that Potala Palace?



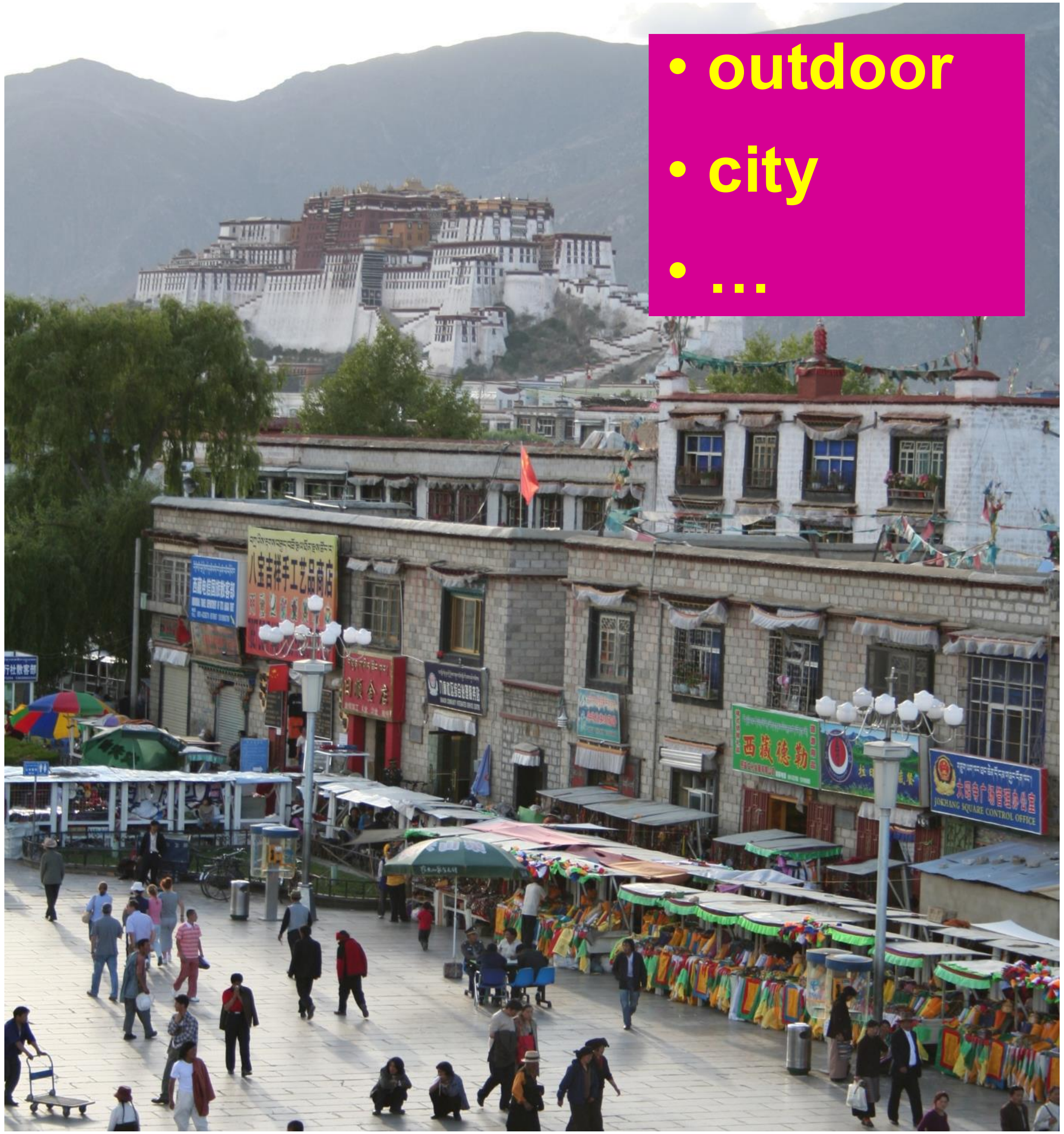
What is “Recognition”?

- Verification: is that a lamp?
- Detection: where are the people?
- Identification: is that Potala Palace?
- Classification: what objects are present?



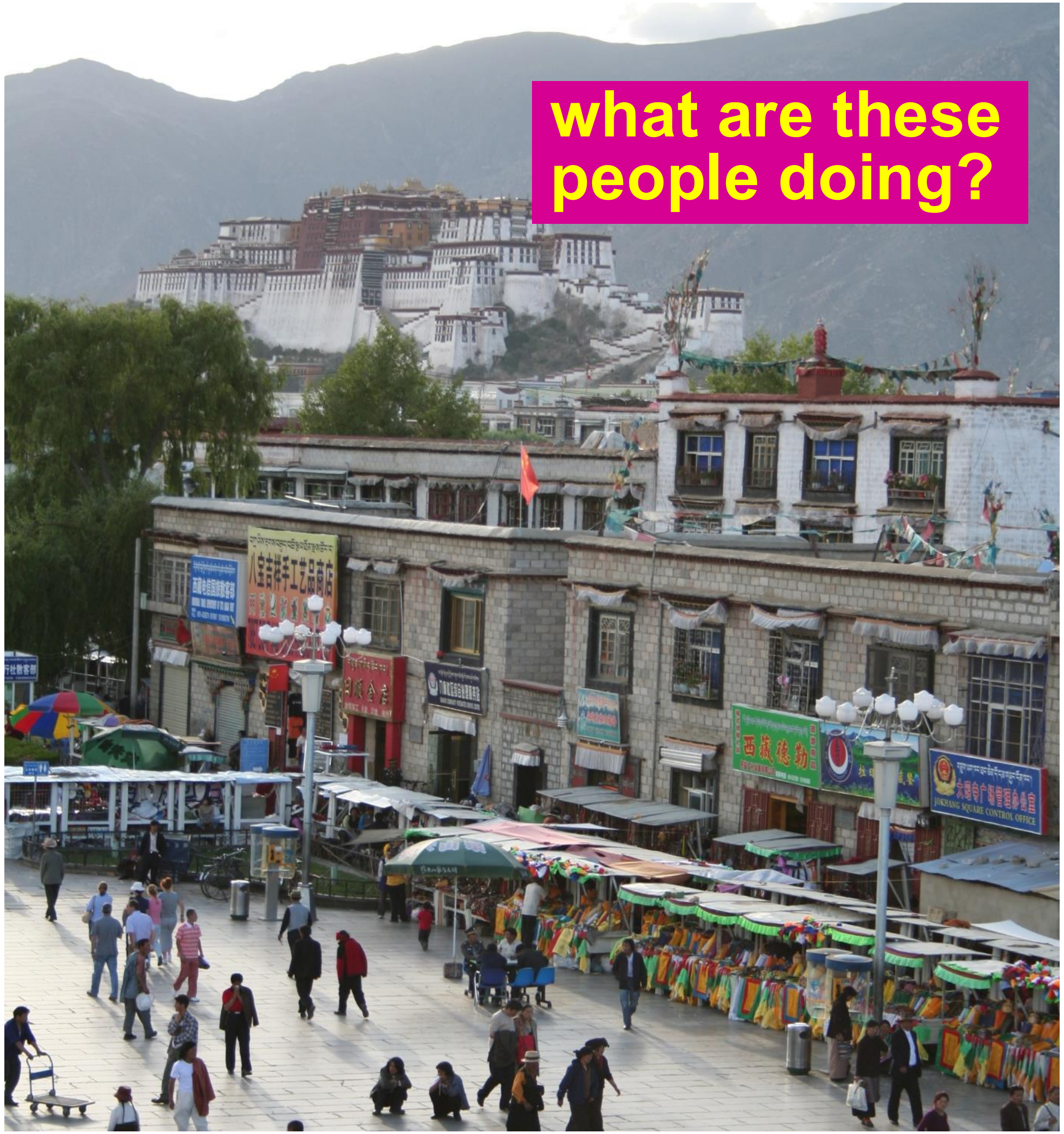
What is "Recognition"?

- Verification: is that a lamp?
- Detection: where are the people?
- Identification: is that Potala Palace?
- Classification: what objects are present?
- Scene and context categorization



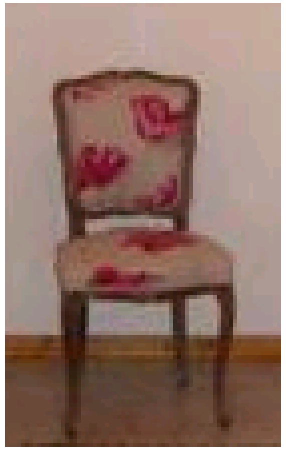
What is "Recognition"?

- Verification: is that a lamp?
- Detection: where are the people?
- Identification: is that Potala Palace?
- Classification: what objects are present?
- Scene and context categorization
- Activity / Event Recognition



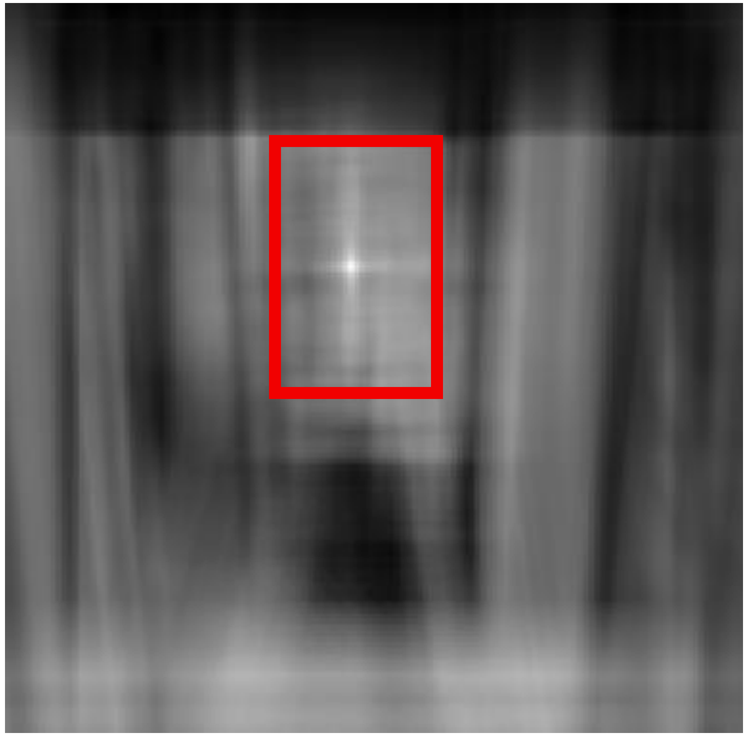
Object recognition: Is it really so hard?

This is a chair



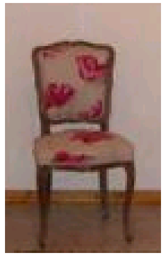
Find the chair in this image

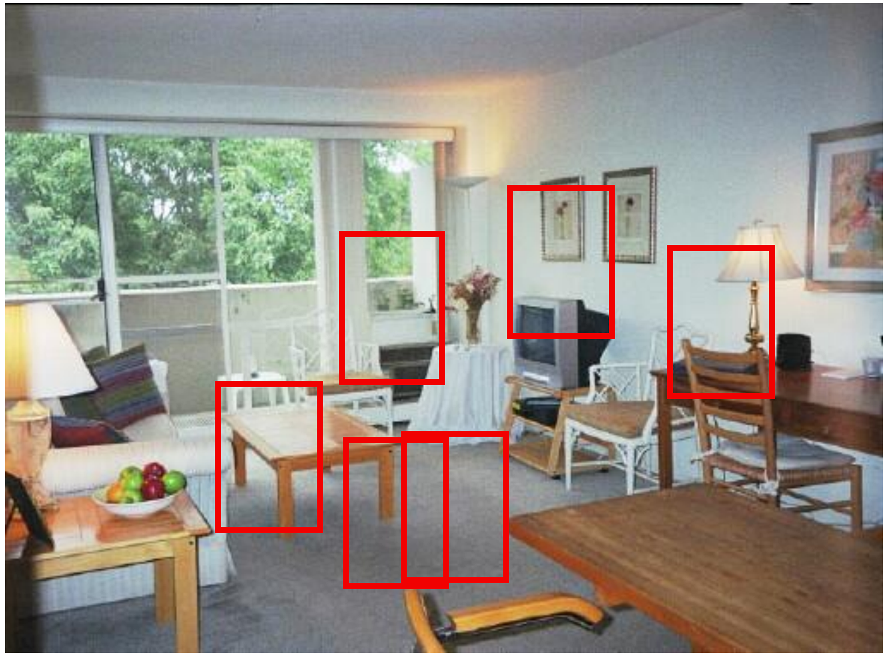
Output of normalized correlation

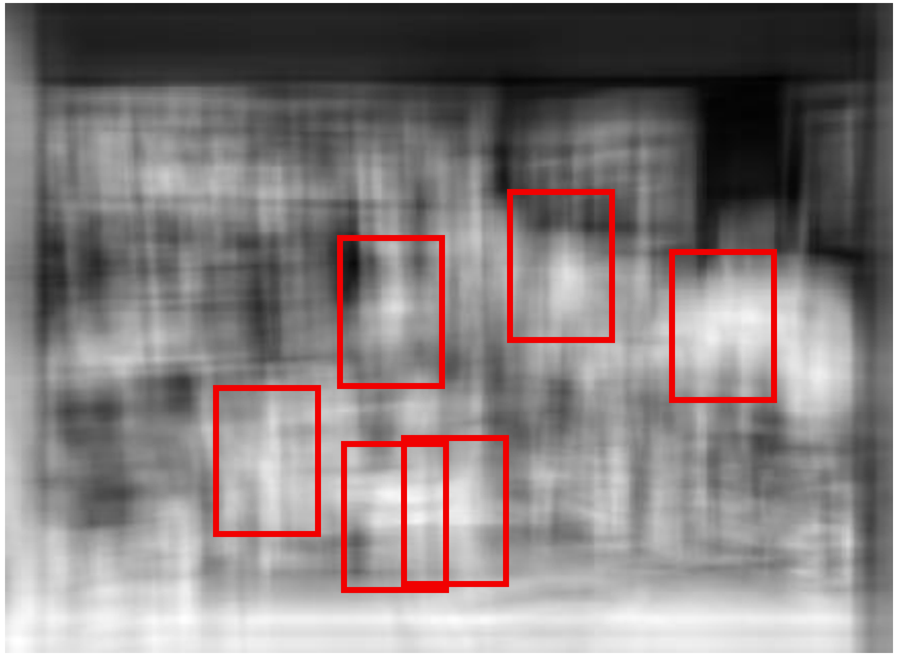


Object recognition: Is it really so hard?

Find the chair in this image



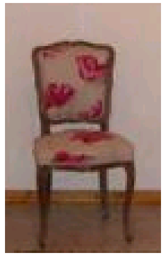




Pretty much garbage:
Simple template matching is not
going to do the trick

Object recognition: Is it really so hard?

Find the chair in this image



A “popular method is that of template matching, by point to point correlation of a model pattern with the image pattern. These techniques are inadequate for three-dimensional scene analysis for many reasons, such as occlusion, changes in viewing angle, and articulation of parts.” Nivatia & Binford, 1977.

Why not use SIFT matching for everything?

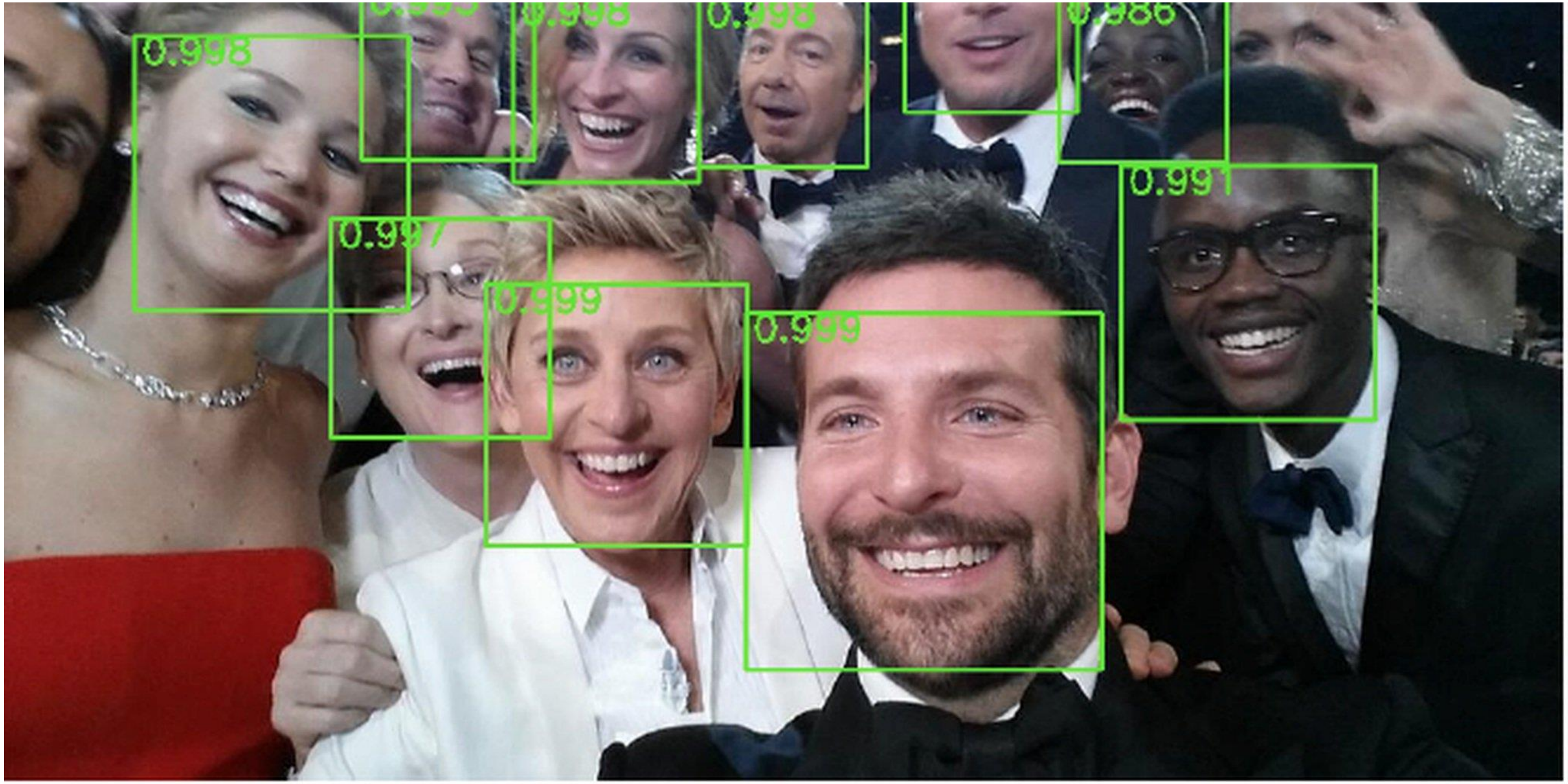
- Works well for object *instances* (or distinctive images such as logos)

- Not great for generic object *categories*

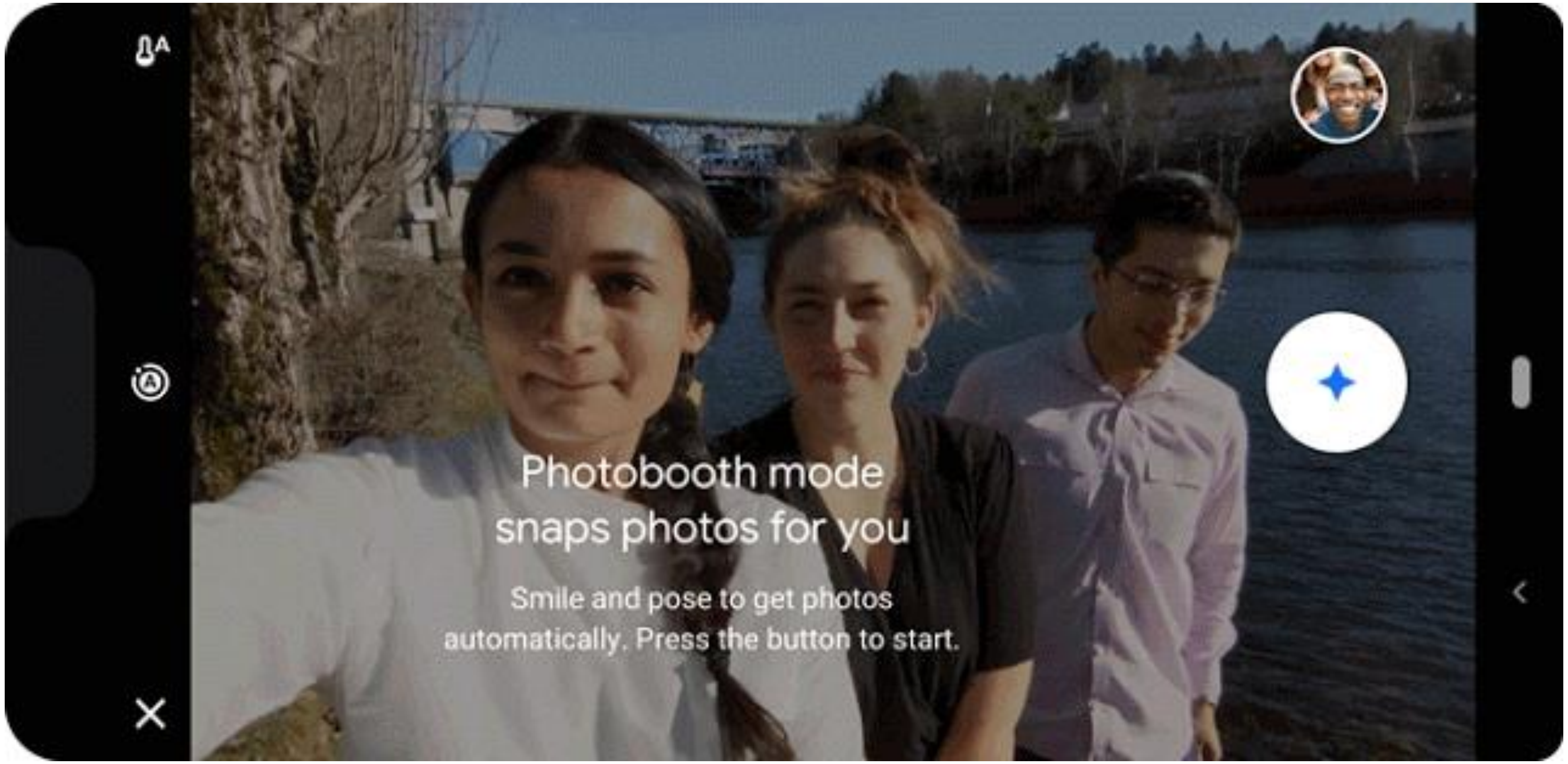
And it can get a lot harder

Brady, M. J., & Kersten, D. (2003). Bootstrapped learning of novel objects. J Vis, 3(6), 413-422

Applications: Photography



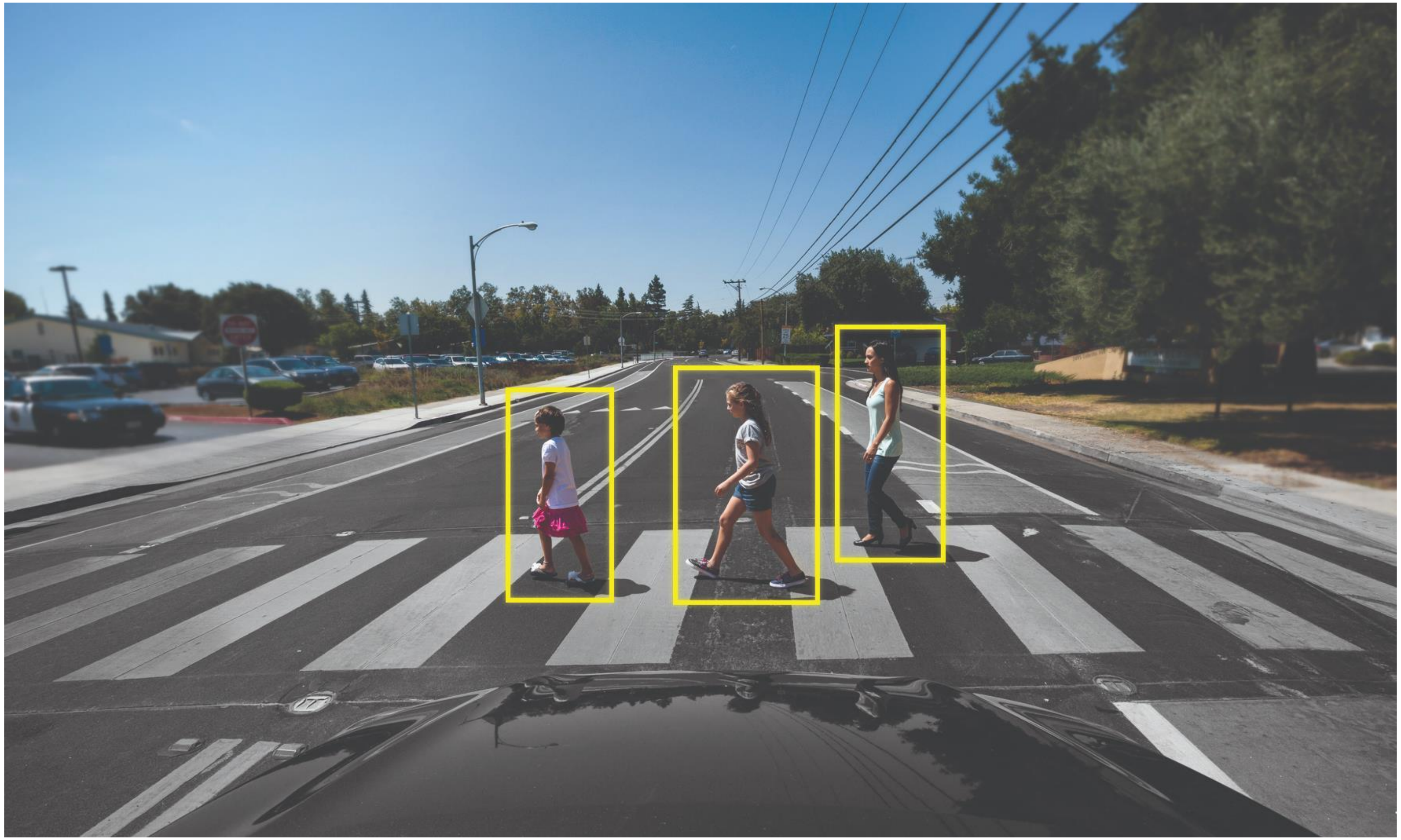
Applications: Shutter-free Photography



Take Your Best Selfie Automatically, with Photobooth on Pixel 3

https://ai.googleblog.com/2019/04/take-your-best-selfie-automatically.html
(Also features "kiss detection")

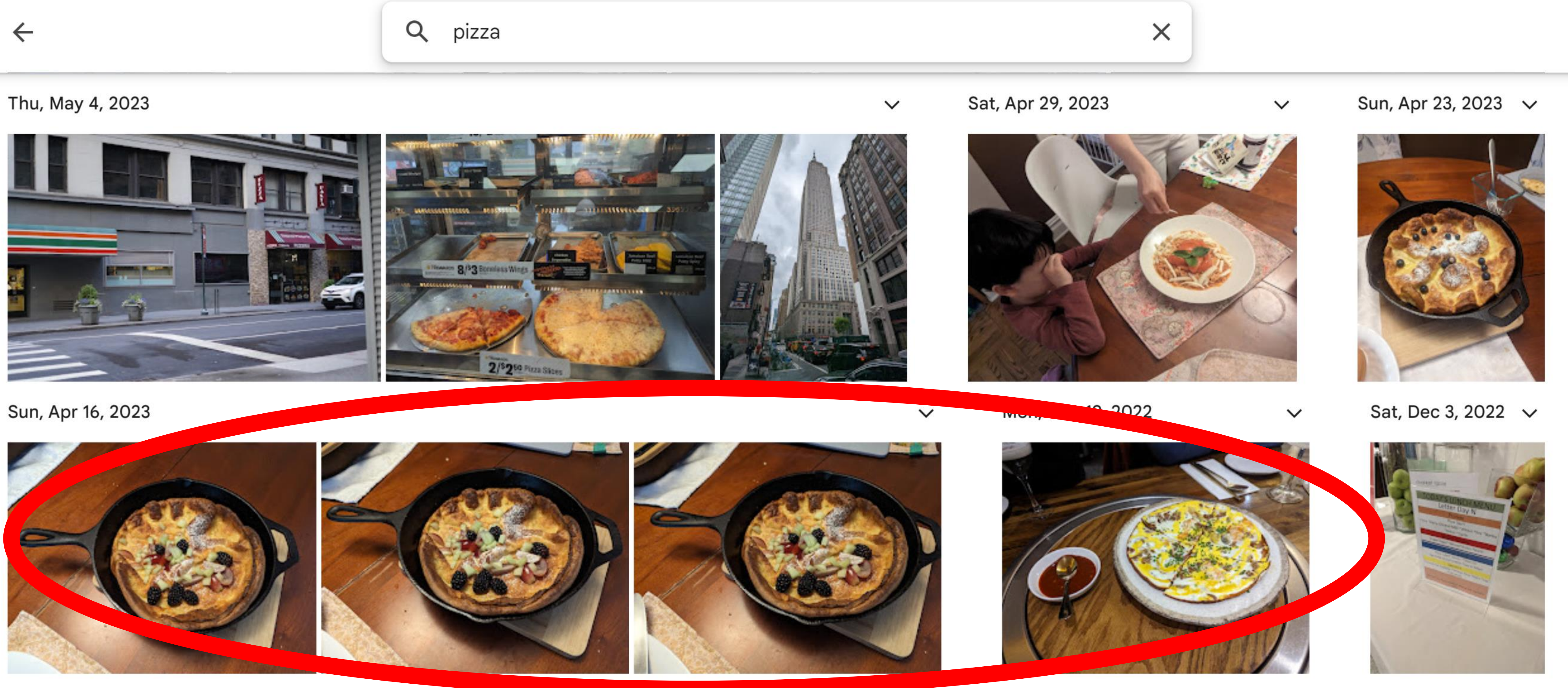
Applications: Assisted / autonomous driving



Applications: Robotics

https://arc.cs.princeton.edu/

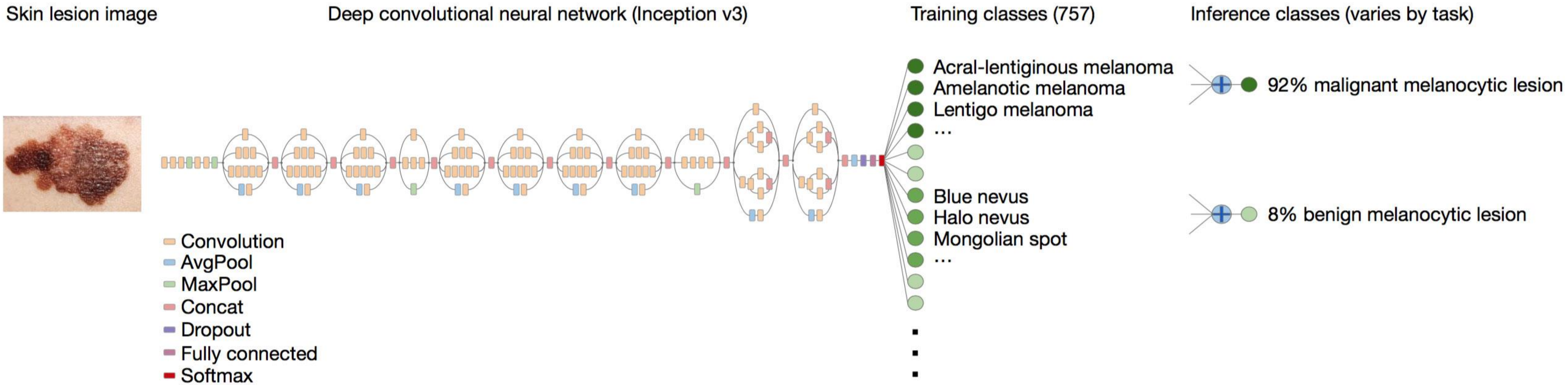
Applications: Photo organization



Source: Google Photos

Not Pizzas!

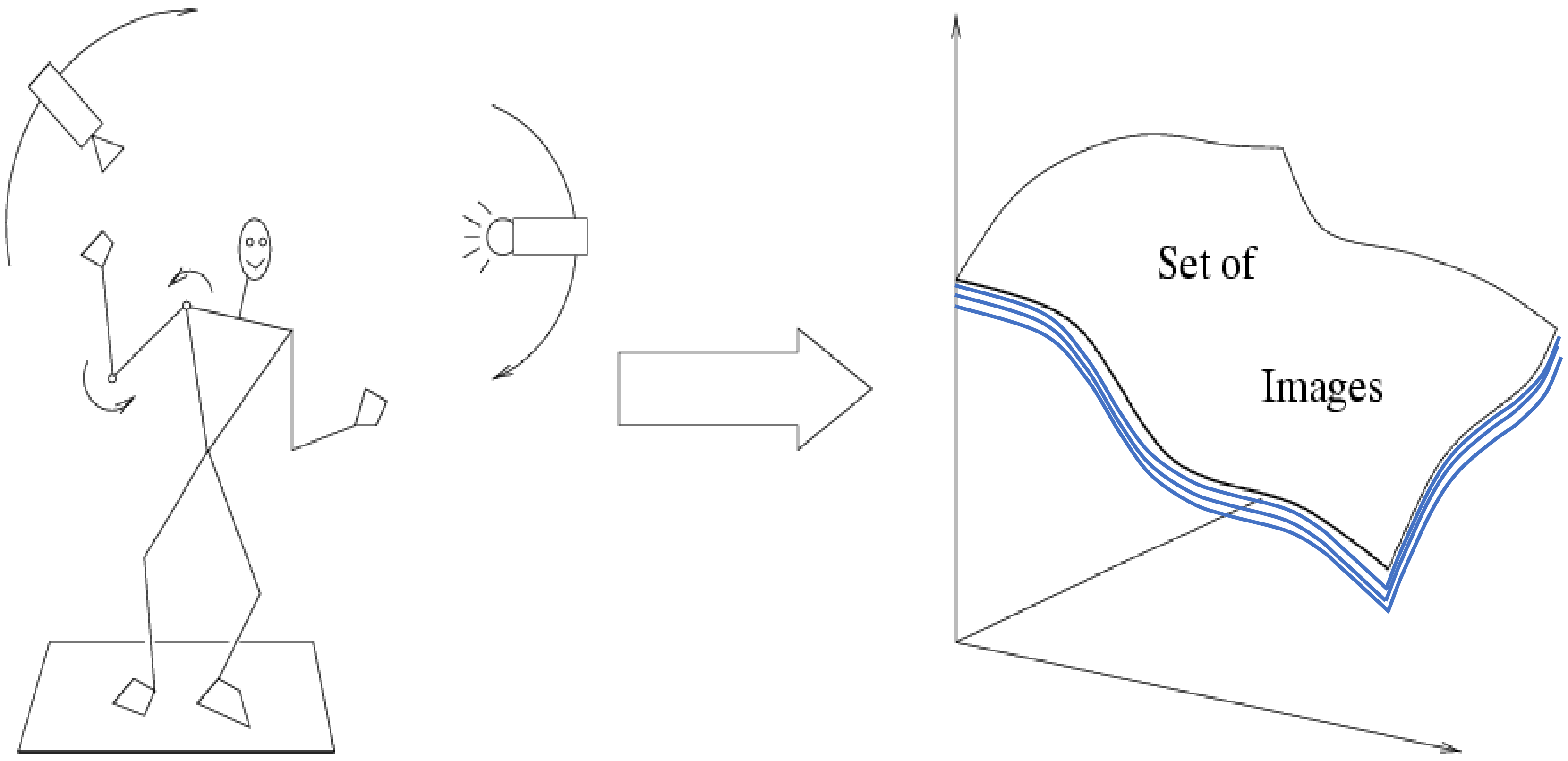
Applications: medical imaging



Dermatologist-level classification of skin cancer

https://cs.stanford.edu/people/esteva/nature/

Why is recognition hard?



Variability: Camera position,
Illumination,
Shape,
etc…

Svetlana Lazebnik

Challenge: lots of potential classes

Challenge: variable viewpoint

Michelangelo 1475-1564

Challenge: variable illumination

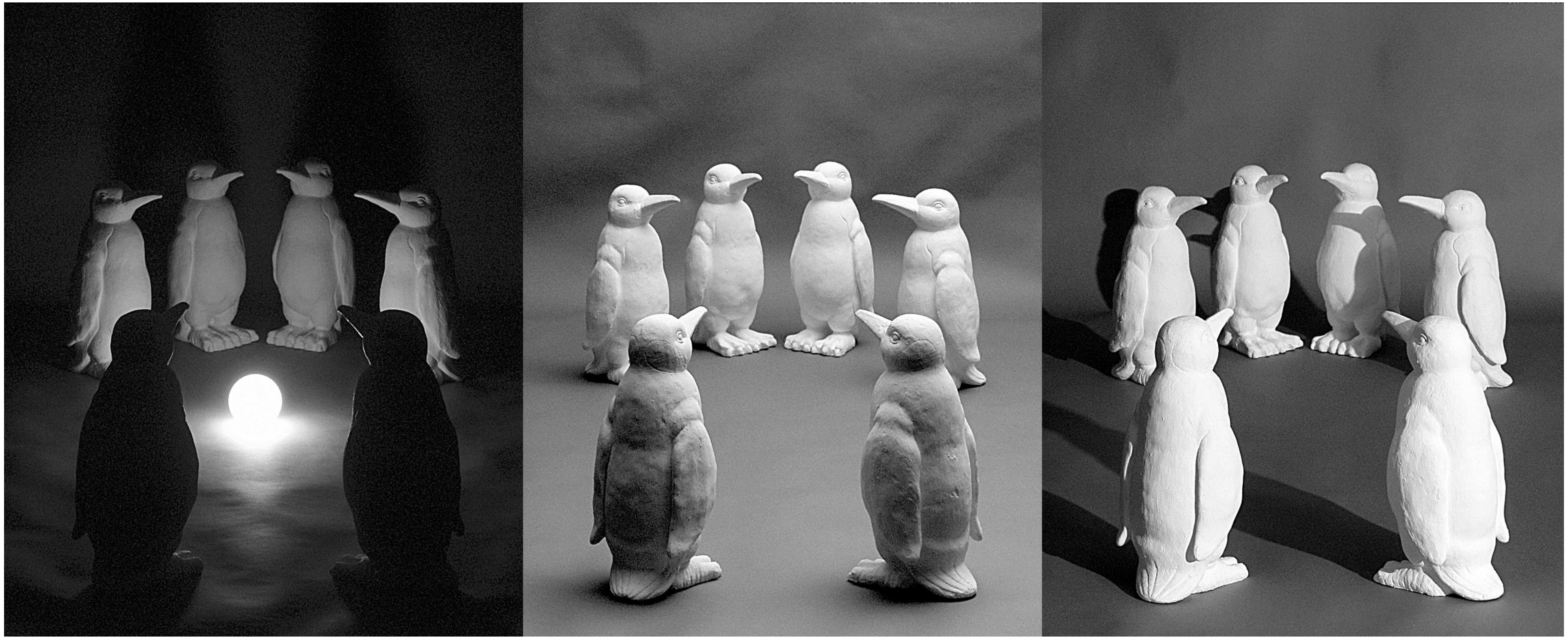
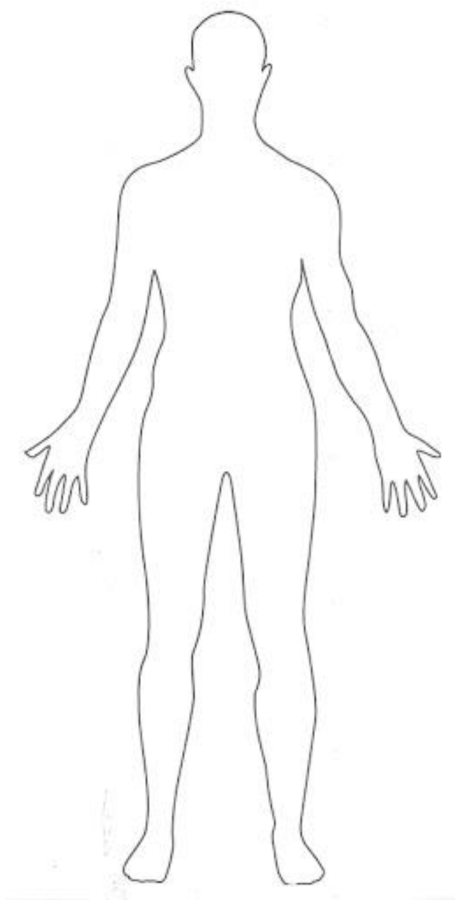


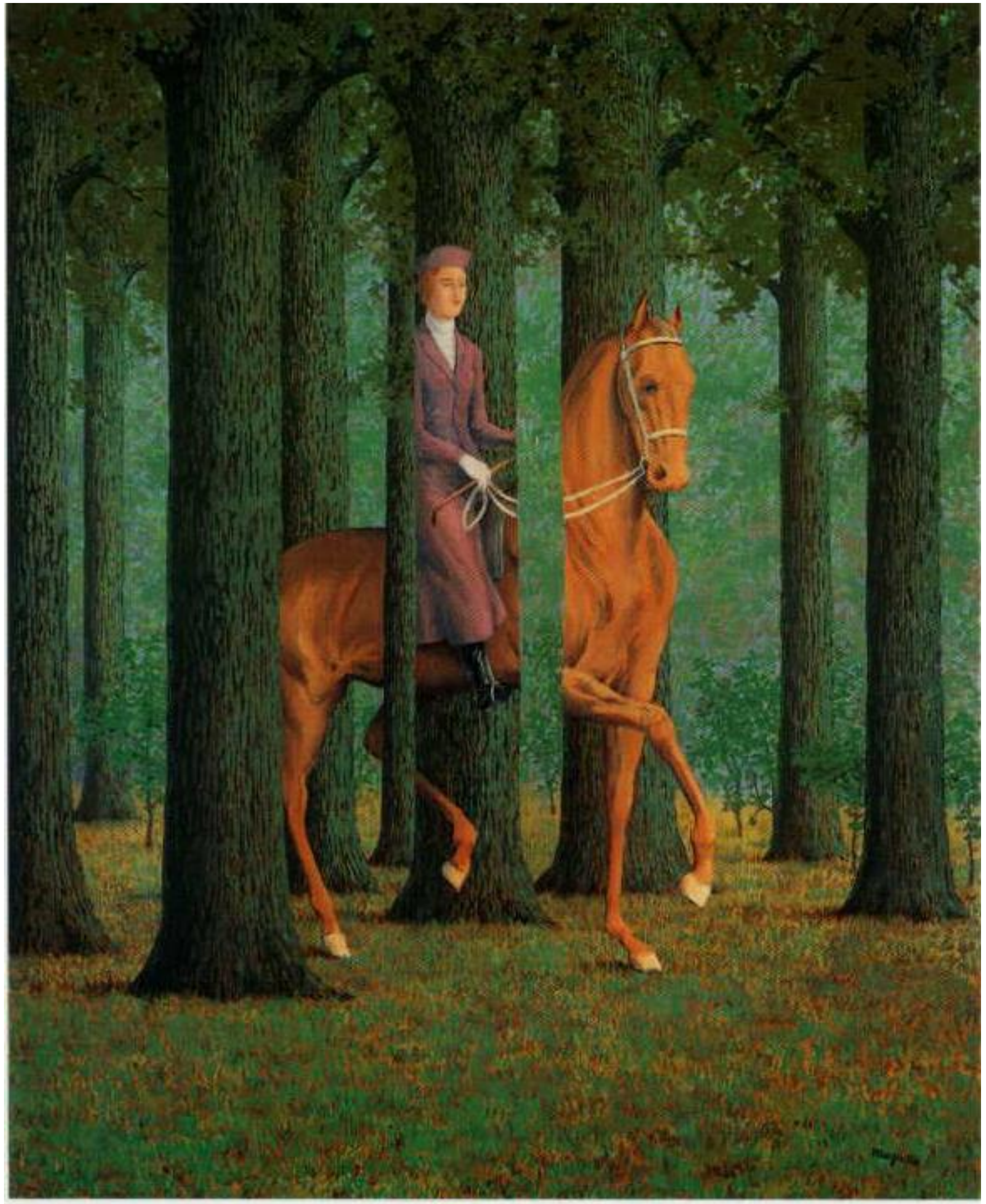
image credit: J. Koenderink

Challenge: scale

Challenge: deformation



Challenge: Occlusion



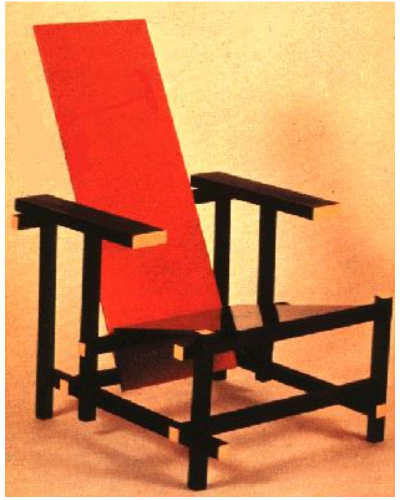
Magritte, 1957

Challenge: background clutter

Kilmeny Niland.
1995

Challenge: intra-class variations

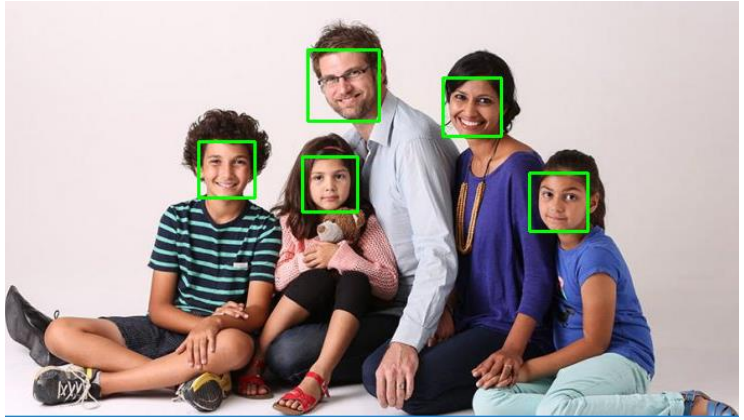


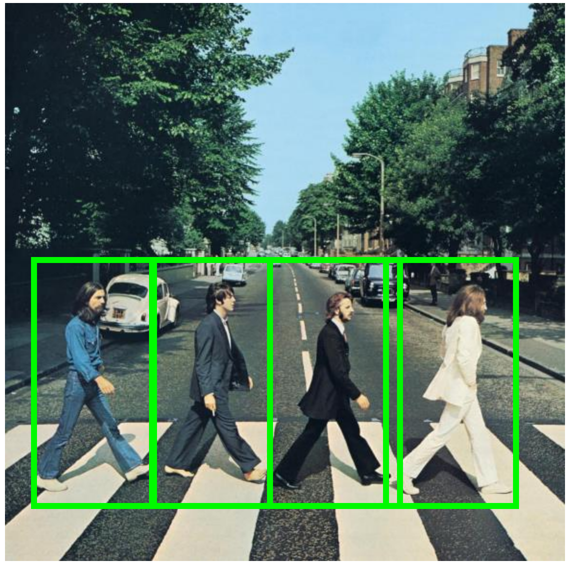



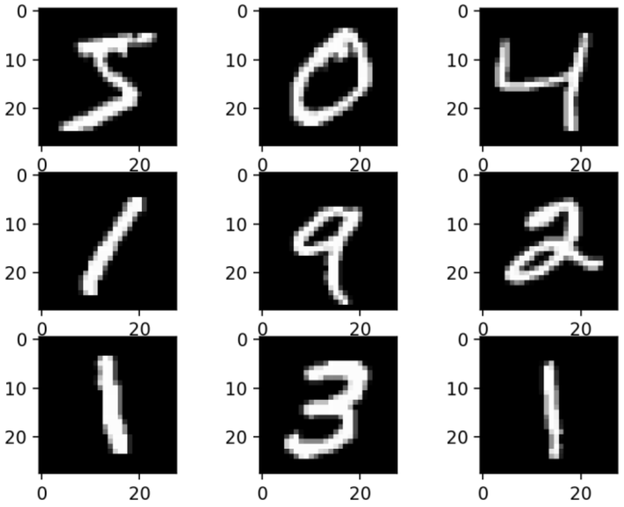
Svetlana Lazebnik

A brief history of image recognition

- What worked in 2011 (pre-deep-learning era in computer vision)
  - Optical character recognition
  - Face detection
  - Instance-level recognition (what logo is this?)
  - Pedestrian detection (sort of)
  - … that’s about it







A brief history of image recognition

- What works now, post-2012 (deep learning era and beyond)
  - Robust object classification across thousands of object categories (rivalling human capabilities)

“Spotted salamander”

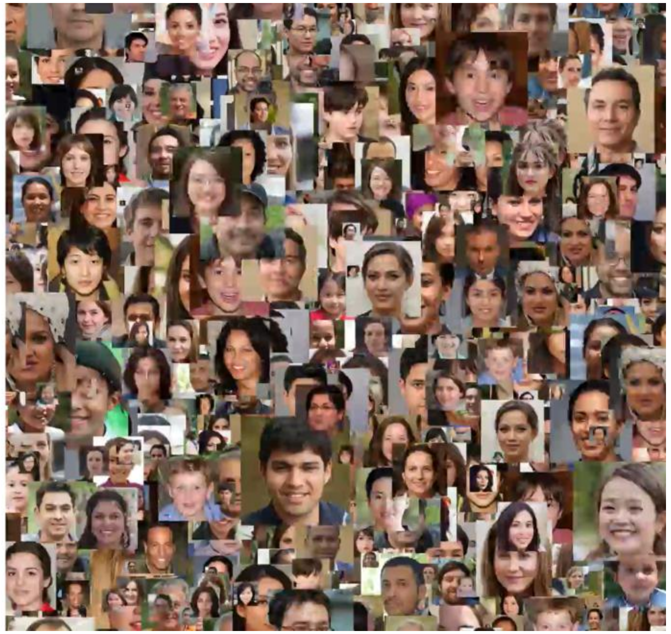
A brief history of image recognition

- What works now, post-2012 (deep learning era and beyond)
    - Face recognition at scale

The New York Times   Account

The Secretive Company That Might End Privacy as We Know It

A little-known start-up helps law enforcement match photos of unknown people to their online images — and "might lead to a dystopian future or something," a backer says.



https://www.nytimes.com/2020/01/18/technology/clearview-privacy-facial-recognition.html

FaceNet: A Unified Embedding for Face Recognition and Clustering

Florian Schroff
fschroff@google.com
Google Inc.

Dmitry Kalenichenko
dkalenichenko@google.com
Google Inc.

James Philbin
jphilbin@google.com
Google Inc.

FaceNet, CVPR 2015

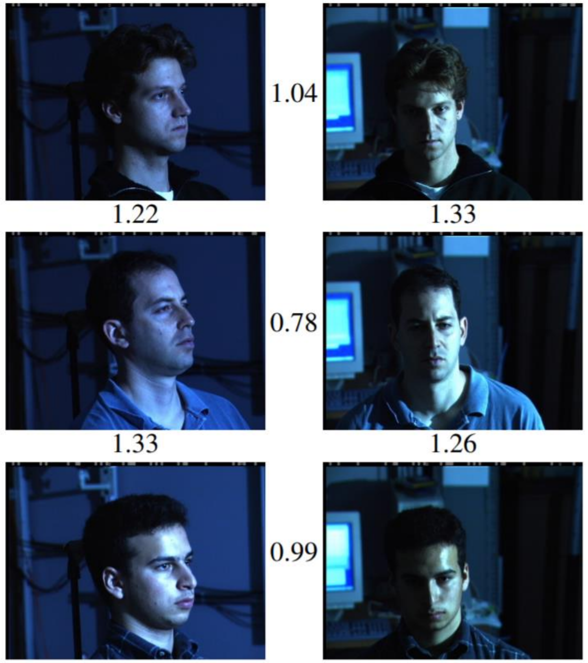


Figure 1. **Illumination and Pose invariance.** Pose and illumination have been a long standing problem in face recognition. This figure shows the output distances of FaceNet between pairs of faces of the same and a different person in different pose and illumination combinations. A distance of 0.0 means the faces are identical, 4.0 corresponds to the opposite spectrum, two different identities. You can see that a threshold of 1.1 would classify every pair correctly.

A brief history of image recognition

- What works now, post-2012 (deep learning era and beyond)
  - High-quality image/video synthesis

A Style-Based Generator Architecture for Generative Adversarial Networks
Tero Karras (NVIDIA), Samuli Laine (NVIDIA), Timo Aila (NVIDIA)
http://stylegan.xyz/paper

These people are not real – they were produced by our generator that allows control over different aspects of the image.

A brief history of image recognition

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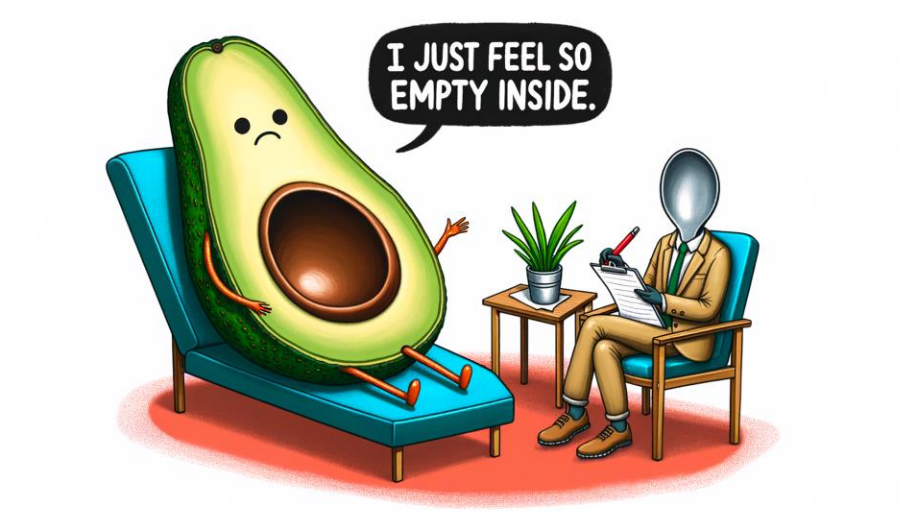
DALL-E 3



An illustration of an avocado sitting in a therapist's chair, saying 'I just feel so empty inside' with a pit-sized hole in its center. The therapist, a spoon, scribbles notes.

Several giant wooly mammoths approach treading through a snowy meadow, their long wooly fur lightly blows in the wind as they walk, snow covered trees and dramatic snow capped mountains in the distance...

Societal impacts

- Privacy invasion (e.g., face/person recognition, biometrics)
- Bias in AI methods (e.g., recognition systems that perform worse on certain demographics)
- Bias in training data (e.g., used to learn or perpetuate biased associations)
- Sources of training data (copyright issues, consent issues, etc.)
- Generative media (e.g., deepfakes, disinformation)
- …

What Matters in Recognition?

- Learning Techniques
  - E.g. choice of classifier or inference method
- Representation
  - Low level: SIFT, HoG, GIST, edges
  - Mid level: Bag of words, sliding window, deformable model
  - Deep learned features
  - Latent diffusion models
- Data
  - More is always better (as long as it is good data)
  - Annotation (labeling data) has historically been a key challenge
  - Now we are seeing powerful models trained from more noisy labels

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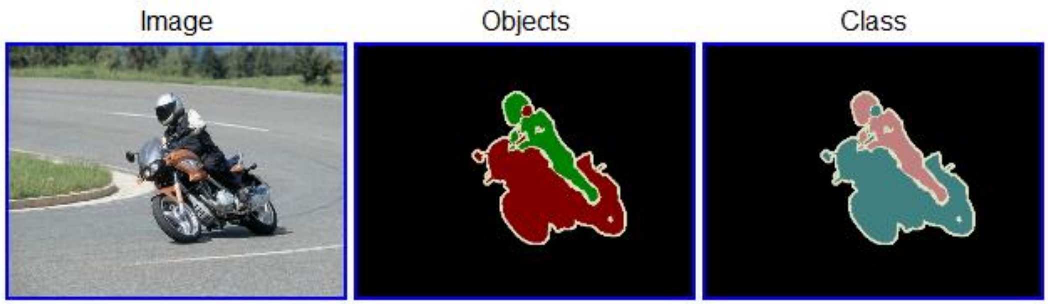
Datasets

- PASCAL VOC [2005-2012]
  - *Not* Crowdsourced, bounding boxes, 20 categories
- CIFAR-10 [2009]
  - 60000 32x32 color images in 10 classes (6000 images per class)
- ImageNet [2010 – current]
  - Huge, Crowdsourced, Hierarchical, *Iconic* objects
- COCO (Common Objects in Context) [2014 – current]
  - Crowdsourced, large-scale objects
- LAION 5B [2022 – current]
  - 5.85 billion noisy image-text pairs

The PASCAL Visual Object Classes Challenge 2009 (VOC2009)

- 20 object categories (aeroplane to TV/monitor)

- Three challenges:
  - Classification challenge (is there an X in this image?)
  - Detection challenge (draw a box around every X)
  - Segmentation challenge (which class is each pixel?)



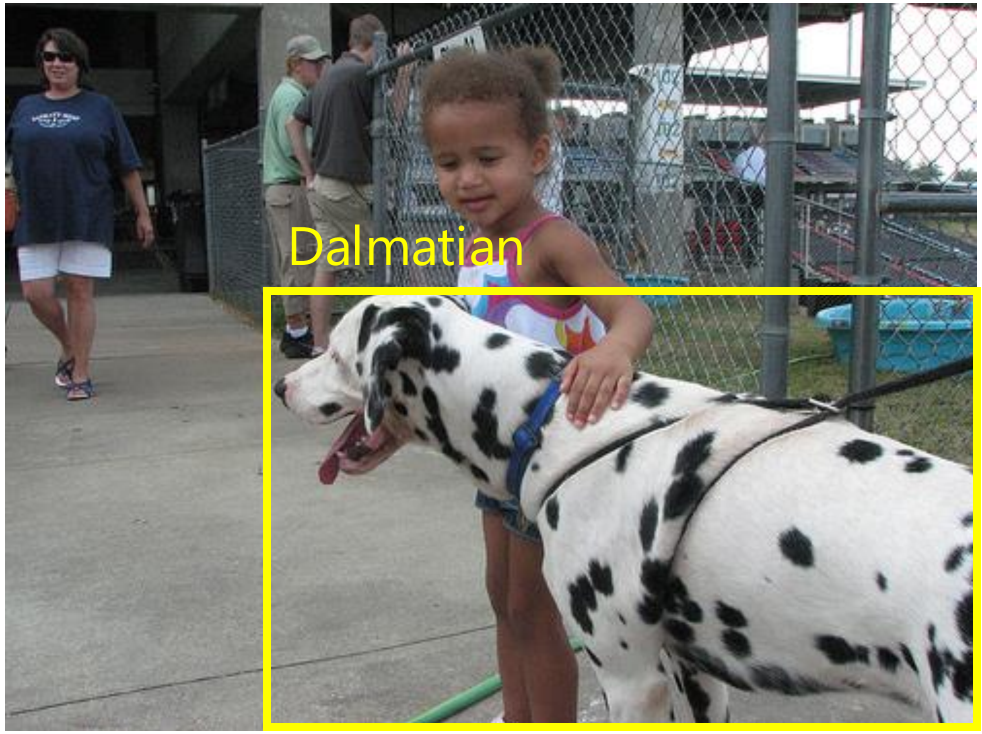
Example segmentation

ImageNet Large Scale Visual Recognition Challenge (ILSVRC)

2010-2017

~~20 object classes~~ ~~22,591 images~~

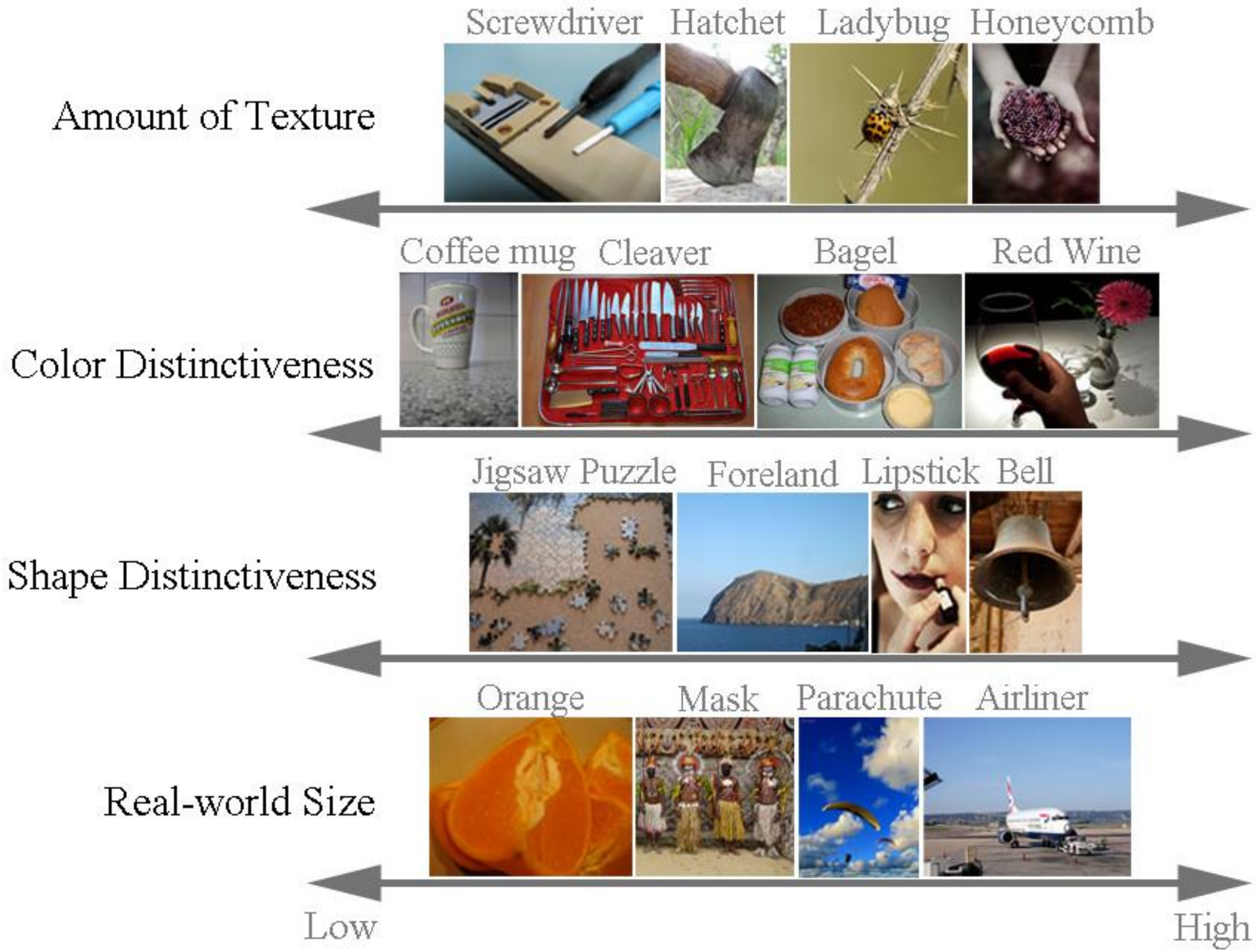
1000 object classes **1,431,167 images**



http://image-net.org/challenges/LSVRC/{2010,2011,2012}

Variety of object classes in ILSVRC

Variety of object classes in ImageNet



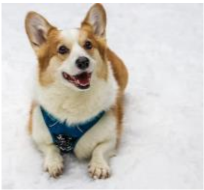
Questions?

Recap: Image classification

- Input: an image
  - Assume it is of fixed size, e.g. 128x128 pixels
- Output: the class label for that image
- Label is generally a discrete label from a pre-defined set
  - e.g. {cat, dog, cow, toaster, apple, tomato, truck, … }

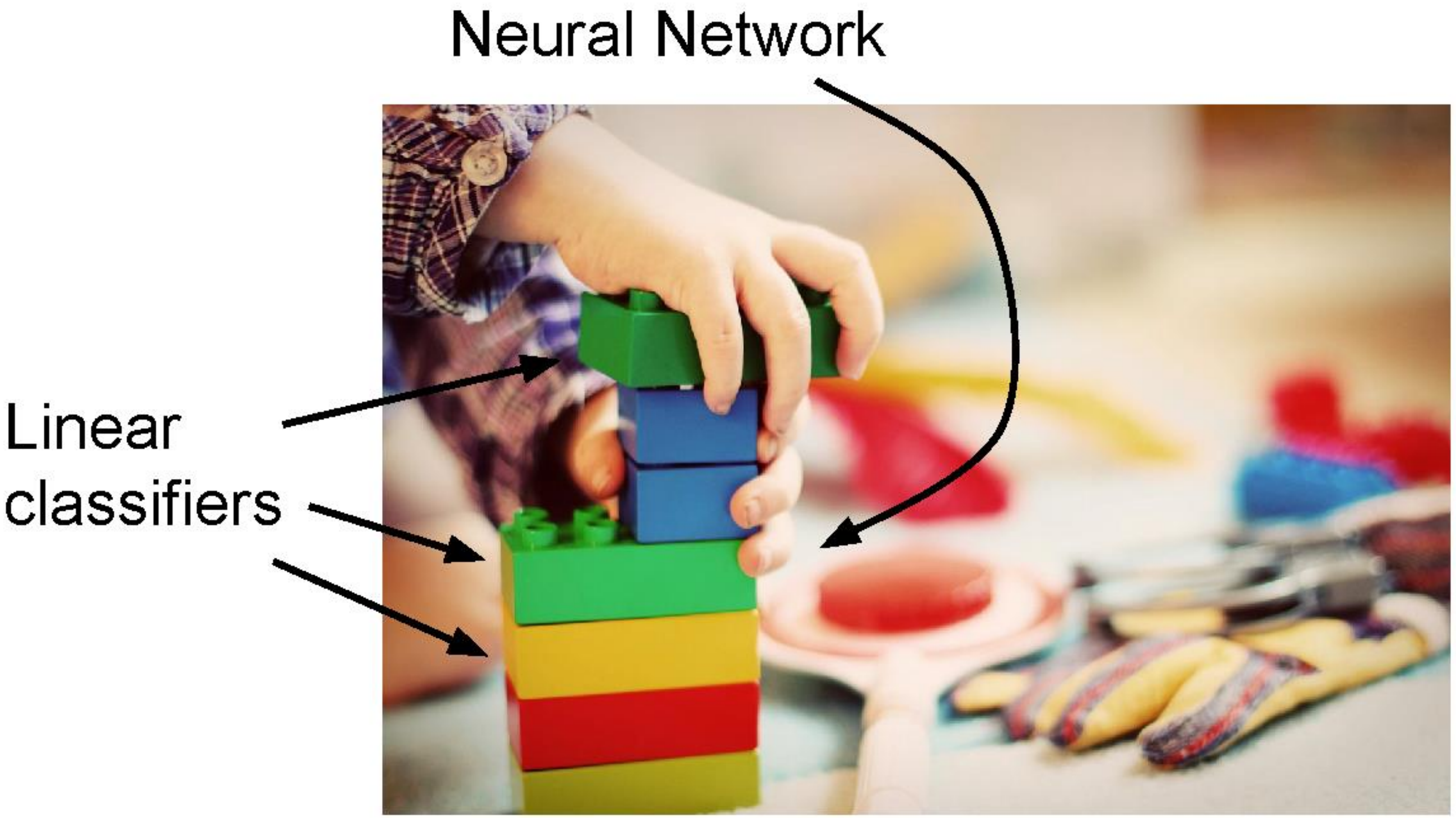
```
def classifier(image):
    //Do some stuff
    return class_label;
```

$f($  $) =$ "Cat"

$f($  $) =$ "Dog"

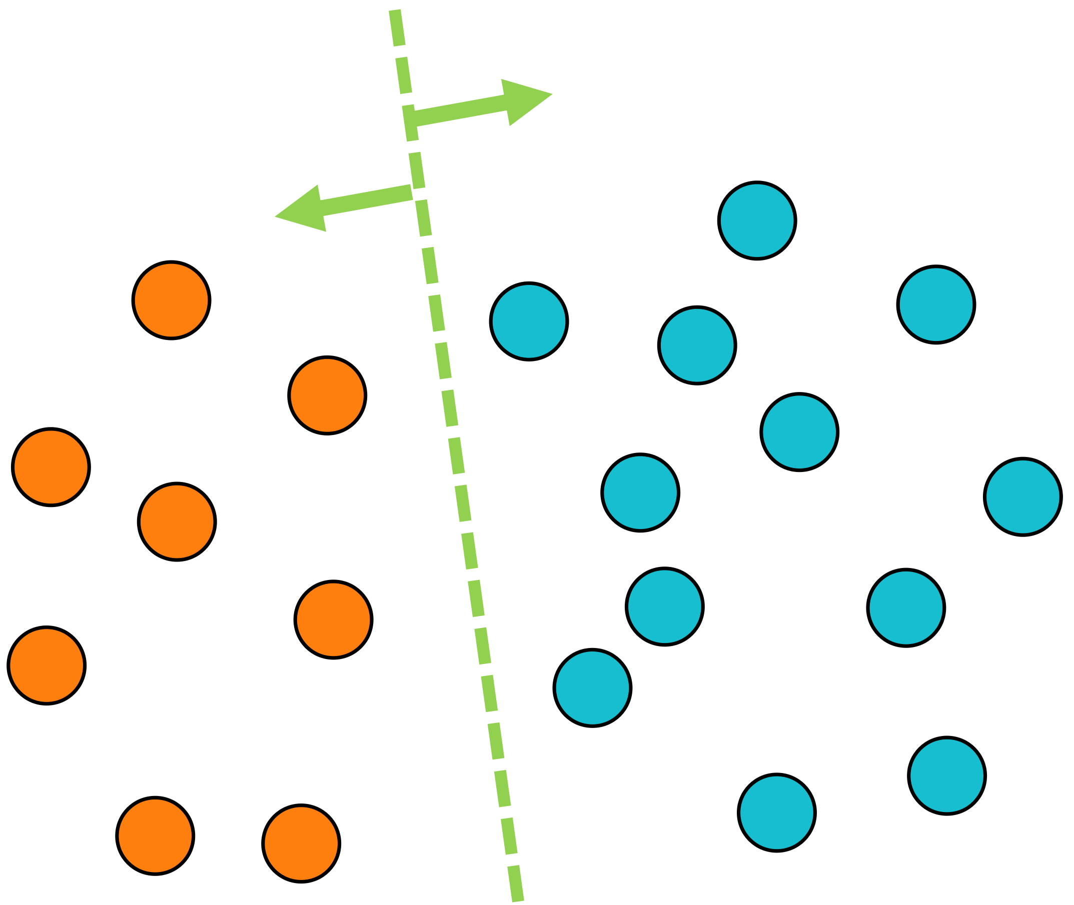
$f($  $) =$ "Toaster"

Linear Classifiers



Linear Classification

- Linear Classifier
  - Store hyperplanes that best separate different classes
  - We can compute continuous class score by calculating (signed) distance from hyperplane



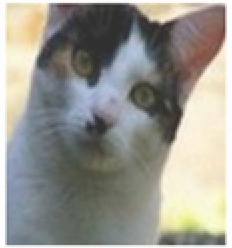
We can interpret this as a linear "score function" for each class.

Linear classifier

$$\boxed{f(x,W)} = \boxed{W}\boxed{x} \quad \boxed{(+b)}$$

3072x1 10x1

10x1 10x3072



$[32x32x3] = 3072$
array of numbers 0...1

10 numbers, indicating class scores

parameters, or “weights”

Slide adapted from Andrej Karpathy and Fei-Fei Li http://vision.stanford.edu/teaching/cs231n/

Softmax classifier

- Interpret Scores as unnormalized log probabilities of classes

$f(x_i, W) = W x_i$ (score function)

$\boxed{\dfrac{e^{f_{y_i}}}{\sum_j e^{f_j}}}$ softmax function

Squashes values into *probabilities* ranging from 0 to 1

$$P(y_i \mid x_i; W)$$

Example with three classes:

$$[1, -2, 0] \rightarrow [e^1, e^{-2}, e^0] = [2.71, 0.14, 1] \rightarrow [0.7, 0.04, 0.26]$$

scores — exponentiated scores — normalized probabilities

Putting it all together

- What we have: a score function and loss function
  - Score function maps an input data instance (e.g., an image) to a vector of scores, one for each category
  - Our current score function is based on linear classifier

$$f(x, W) = Wx + b$$

f: score function
x: input instance
W, b: parameters of a linear (actually affine) function

- Find **W** and **b** that minimize a *loss* over labeled training data, e.g. cross-entropy loss

$$L = \frac{1}{N} \sum_i -\log\left(\frac{e^{f_{y_i}}}{\sum_j e^{f_j}}\right)$$

Beyond linear classifiers: Neural networks

(**Before**) Linear score function: $f = Wx$

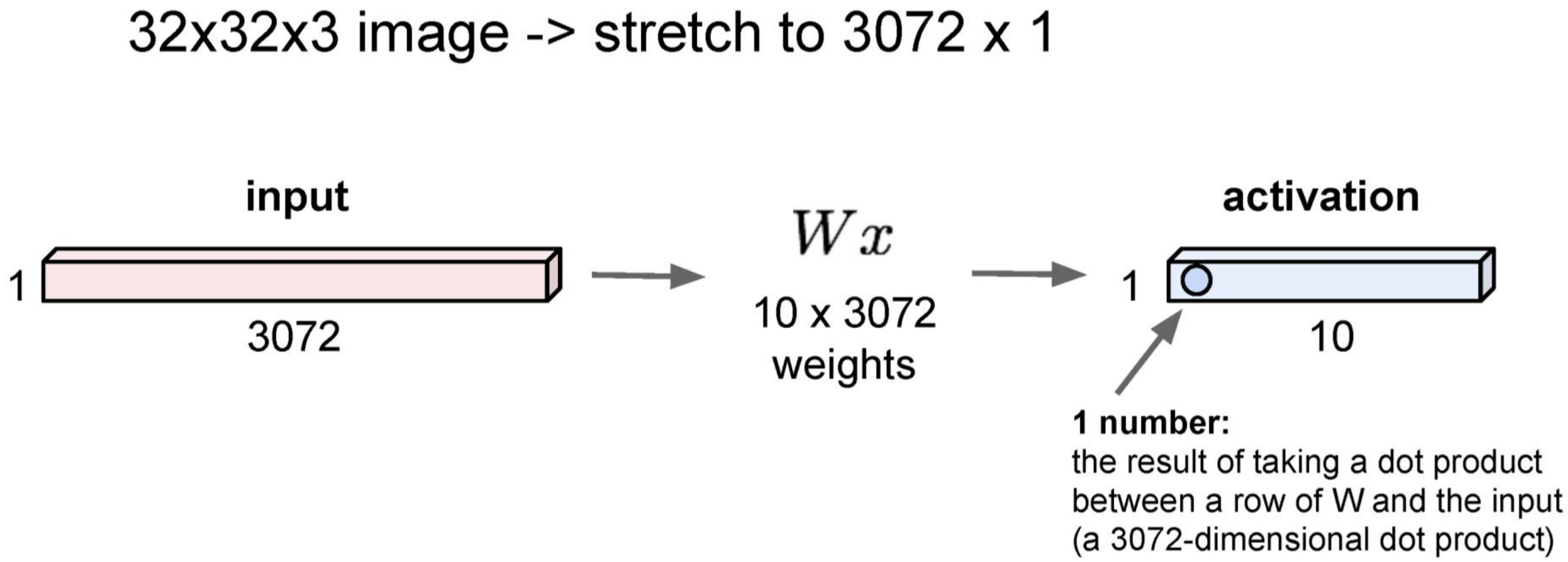
(**Now**) 2-layer Neural Network $f = W_2 \max(0, W_1 x)$
or 3-layer Neural Network

$$f = W_3 \max(0, W_2 \max(0, W_1 x))$$

also called "**Multilayer Perceptrons**" (MLPs)
(used in NeRFs)

Adapted from Fei-Fei Li & Andrej Karpathy & Serena Leu

Fully connected layer



- Just like a linear classifer – but in this case, just one layer of a larger *network*

Adapted from Fei-Fei Li & Andrej Karpathy & Serena Le

Next

- Convolutional Neural Networks (CNNs)

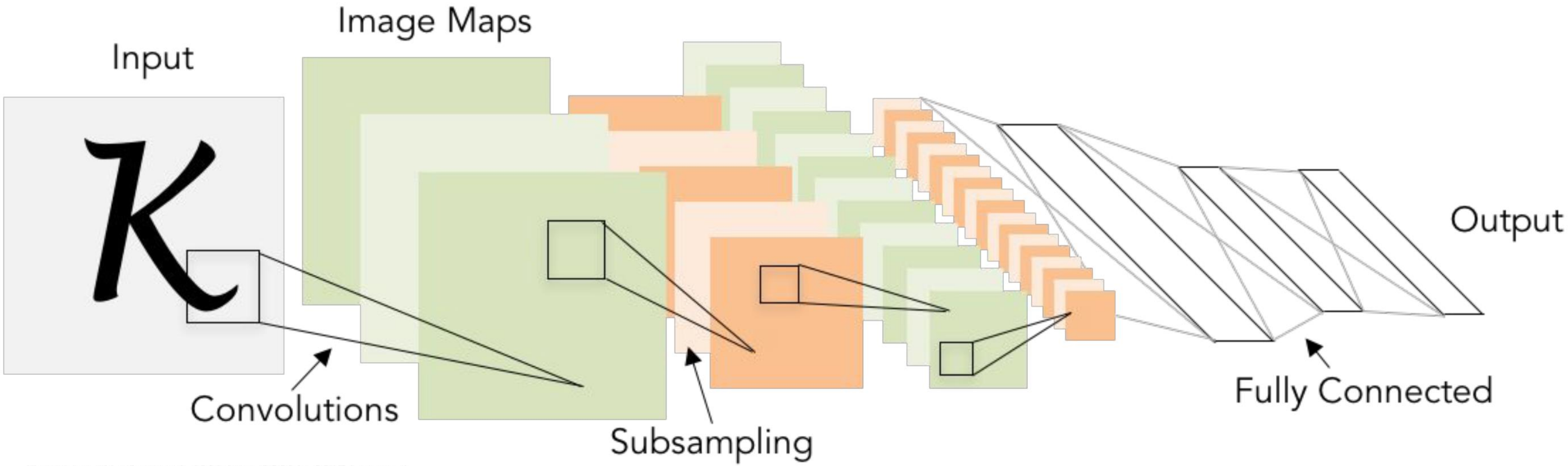


Illustration of LeCun et al. 1998 from CS231n 2017 Lecture 1